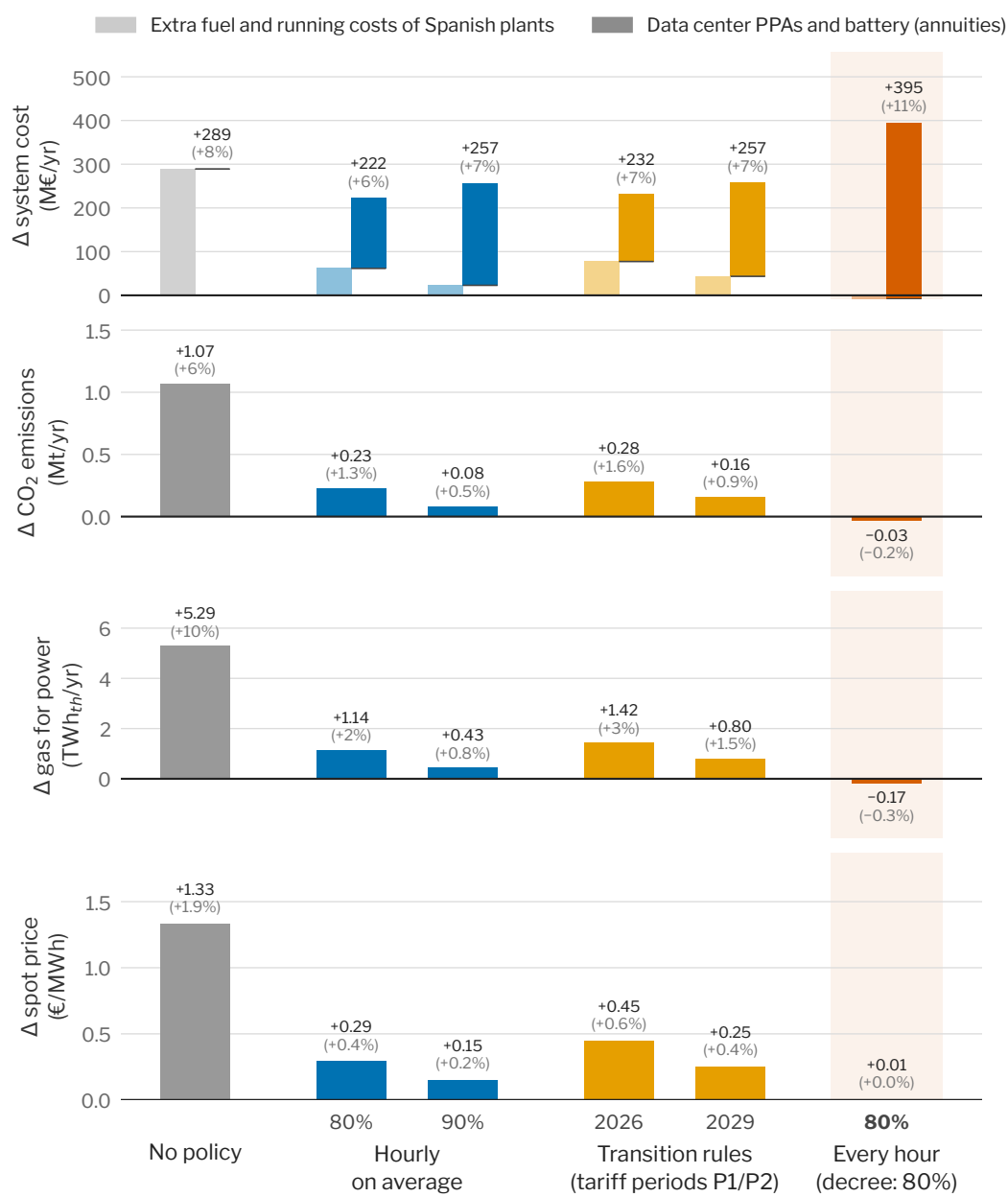

Change in the Spanish power system with a 500 MW data center in Aragon against a baseline without it: each bar carries the change and, in grey, the same change in percent of the baseline level. System cost: fuel, CO2 (EU ETS) and running costs of Spanish power plants and storage (light; imports excluded), then the data center's PPA and battery annuities on top (dark), ending at the total. CO2: fuel burnt in Spanish plants times its carbon content. Gas: fuel burnt in Spanish gas-fired power plants (CCGT, OCGT). Spot price: hourly nodal prices weighted by Spanish demand, the data center excluded. PPAs anywhere in mainland Spain. Transition rules: the revised draft's optional phase-in, a floor applied hour by hour only in peak tariff periods (P1: weekday daytime and evening hours in winter and July; P2: adjoining hours and March/November): 60% in P1 under the 2026 rules, then 70% in P1 and 60% in P2 from 2029. All other hours are unconstrained.